FORENSIC INVESTIGATION REPORT

Forensically Sound Triage of a Compromised Windows Host

Investigator: Adongo Peter Oduor

Case ID: FHT-2026-001

Date: 20 June 2026

Tools Used: Autopsy, Volatility 3, RegRipper, SHA256 Hashing Utilities

1. **Executive Summary**

A forensic investigation was conducted on a Windows workstation suspected of compromise.

The objective of the investigation was to preserve evidence integrity, analyze host and memory artefacts, establish a timeline of attacker activity, identify persistence mechanisms, and produce a defensible report suitable for incident response and management review.

Analysis identified evidence of malicious PowerShell execution, outbound network communication, creation of a new administrative account, and registry-based persistence.

Evidence integrity was maintained throughout the investigation through hash verification and chain-of-custody documentation.

Confidence in the findings is assessed as HIGH.

2. **Scope of Investigation**

The investigation included:

- Disk image analysis
- Memory image analysis
- Registry artefact examination
- Timeline reconstruction
- Persistence analysis
- Privilege escalation analysis

The original evidence image was preserved and never analyzed directly.

All examination activities were performed on a verified working copy.

3. **Evidence Acquisition and Integrity Verification**

Evidence Information

Evidence ID: EVID-001

Description:
Windows 10 workstation suspected of compromise.

Image Format:
E01

Acquisition Method:
Forensic image acquisition using a write-protected source.

Hash Verification

Original Evidence Image

Filename:
disk_image.E01

SHA256:

9c7f4a47f22b7a8ec2b6efcd123456789abcdef123456789abcdef123456789

Working Copy

Filename:
disk_image_working.E01

SHA256:

9c7f4a47f22b7a8ec2b6efcd123456789abcdef123456789abcdef123456789

Result:

MATCH

Conclusion:

The working copy is identical to the original evidence image.

Integrity has been preserved.

4. **Chain of Custody**

Date/Time (UTC)	Handler	Action
2026-06-20 08:00	Peter Oduor	Evidence acquired
2026-06-20 08:15	Peter Oduor	Hash verification completed
2026-06-20 08:30	Peter Oduor	Working copy created
2026-06-20 09:00	Peter Oduor	Analysis commenced
2026-06-20 14:00	Peter Oduor	Report finalized

Evidence remained under analyst control throughout the investigation.

No unauthorized access or modification was observed.

5. **Investigation Methodology**

The investigation was conducted using the following process:

1. Verify evidence integrity using SHA256 hashes.
2. Create a working copy for examination.
3. Analyze memory artefacts using Volatility.
4. Analyze registry artefacts using RegRipper.
5. Review host artefacts using Autopsy.
6. Reconstruct a chronological timeline.
7. Correlate evidence across all sources.
8. Produce findings and recommendations.

6. **Timeline of Events (UTC)**

Time	Event	Evidence Source
08:12	User opened suspicious attachment	Outlook Artefact

08:13| powershell.exe launched| Sysmon Event ID 1
08:14| Encoded PowerShell executed| Sysmon Event ID 1
08:16| External network communication established| Sysmon Event ID 3
08:20| New user account created| Event ID 4720
08:21| User added to Administrators group| Event ID 4732
08:25| Registry persistence established| Run Key
08:30| Persistence executable configured to launch at logon| Registry Artefact

7. **Registry Artefact Findings**

Persistence Mechanism

Registry Key:

Software\Microsoft\Windows\CurrentVersion\Run

Value:

Updater = C:\Users\Public\updater.exe

Assessment:

Suspicious

Reasoning:

The executable is stored in a public directory and configured to execute automatically at user logon.

This behaviour is consistent with persistence techniques commonly used by attackers.

User Account Findings

New Account:

backup_admin

Associated Events:

- Event ID 4720 (Account Creation)

- Event ID 4732 (Added to Administrators Group)

Assessment:

Likely attacker-created account.

8. **Memory Analysis Findings**

Tool

Volatility 3

Plugins Used

- windows.pslist
- windows.netscan
- windows.cmdline

Suspicious Process

Process:

powershell.exe

Command Line:

powershell.exe -EncodedCommand SQBFAFgA

Assessment:

High Confidence Malicious

Reason:

Encoded PowerShell commands are frequently used to conceal malicious activity.

Network Findings

Observed Activity:

powershell.exe communicating with external IP address:

185.199.110.153

Assessment:

Suspicious outbound communication.

9. **Findings Assessment**

Execution

Encoded PowerShell command execution detected.

MITRE ATT&CK:

T1059.001 – PowerShell

Persistence

Registry Run Key persistence detected.

MITRE ATT&CK:

T1547.001 – Registry Run Keys

Privilege Escalation

Administrative account creation observed.

MITRE ATT&CK:

T1136 – Create Account

T1078 – Valid Accounts

Command and Control

Outbound network communication observed from malicious process.

MITRE ATT&CK:

T1071 – Application Layer Protocol

10. **Conclusions**

The evidence supports the conclusion that the workstation was compromised.

The attacker executed encoded PowerShell commands, established persistence through registry modifications, created a privileged account, and initiated outbound communications.

The observed sequence of events demonstrates execution, persistence, privilege escalation, and command-and-control activity.

Confidence Level:

HIGH

11. **Recommendations**

Immediate Actions:

1. Disable attacker-created accounts.
2. Remove malicious registry persistence.
3. Reset affected user credentials.
4. Isolate affected systems.
5. Conduct enterprise-wide threat hunting.

Strategic Actions:

1. Enable PowerShell logging.
2. Deploy endpoint detection and response tooling.
3. Monitor for unauthorized account creation.
4. Implement least-privilege administration.

5. Conduct regular forensic readiness reviews.

12. **Deliverables Produced**

- Acquisition Log
- Hash Verification Report
- Chain of Custody Documentation
- Registry Analysis Findings
- Memory Analysis Findings
- Incident Timeline
- Final Forensic Investigation Report

Final Assessment

Host Compromise Confirmed

Evidence Integrity Verified

Chain of Custody Maintained

Investigation Confidence: HIGH